

Problem 11.16

Calculate k if a deposit of 1 accumulates to 2.7183 in 10 years at a force of interest given by

$$\delta_t = \begin{cases} kt, & 0 < t \leq 5, \\ 0.04kt^2, & t > 5. \end{cases}$$

Solution

For a time-varying force of interest, the accumulation factor over 10 years is

$$a(10) = \exp \left(\int_0^{10} \delta_t dt \right).$$

Because the force of interest changes at time 5, split the integral into two parts:

$$2.7183 = \exp \left(\int_0^5 kt dt + \int_5^{10} 0.04kt^2 dt \right).$$

Evaluate the first integral:

$$\int_0^5 kt dt = \left. \frac{kt^2}{2} \right|_0^5 = \frac{25k}{2}.$$

For the second integral,

$$\int_5^{10} 0.04kt^2 dt = \left. \frac{0.04kt^3}{3} \right|_5^{10} = \frac{0.04k}{3} (10^3 - 5^3).$$

Thus,

$$\int_5^{10} 0.04kt^2 dt = \frac{0.04k}{3} (1000 - 125) = \frac{35k}{3}.$$

The total exponent is therefore

$$\frac{25k}{2} + \frac{35k}{3} = \frac{75k + 70k}{6} = \frac{145k}{6}.$$

Hence,

$$2.7183 = e^{145k/6}.$$

Taking natural logarithms and solving for k gives

$$k = \frac{6 \ln(2.7183)}{145}.$$

Therefore,

$$\boxed{k \approx 0.041379587}.$$